

Elastic storage and recoil dynamics

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The catapult mechanism: Elastic recoil of fascial tissues

During the 1980s, those studying the field of muscular physiology were intrigued by the capacity of kangaroos to perform powerful jumps of up to 13 m in length. Because these animals do not have bulky leg muscles, the general assumption was that their leg musculature would contain some unusual muscle fibers that enabled them to perform explosive contractions.

In fact, some researchers were convinced that they would find “super-fast twitch fibers” in the kangaroos’ hind limbs. However, no matter how hard the researchers looked, they could not find any unusual muscle fibers. This left them puzzled: if muscles create the force for these impressive jumps, why do these animals contain the same muscle fibers as a koala? Eventually, they looked for the answer where nobody had looked previously, in the properties of the tendons. It was here that they found an amazing capacity, which was subsequently called the “catapult effect”. Similar to an elastic spring of stainless steel, the long tendons were able to store and release kinetic energy with amazing efficiency (Kram and Dawson, 1998).

When the kangaroo hits the ground their tendons, as well as the fascial aponeuroses of their hind legs, are tensioned like elastic bands. The subsequent release of this stored energy

is what enables these amazing jumps. Shortly afterwards, the same mechanism was also discovered in gazelles. These animals are capable of impressive leaping as well as running, although their musculature is not particularly strong. Because gazelles are generally considered to be rather delicate, it made the elastic springiness all the more interesting. Similar impressive elastic storage capacities were later confirmed to exist in horses.

Homo sapiens: The elastic “gazelle” within the primate family

It was only when the technology of high-resolution ultrasound examination achieved a sufficiently high level of resolution to observe single sarcomeres in muscles that a similar orchestration of loading between muscle and fascia was discovered in human locomotion. In fact it was found, rather surprisingly, that human fasciae have a similar kinetic storage capacity to that of kangaroos and gazelles (Sawicki et al., 2009). This capacity is not only used when we run or jump but also when walking, as a significant part of the energy of the movement comes from the same elastic springiness of the collagenous tissues described above.

Neither chimpanzees, bonobos, orangutans nor any other primate seems to have developed a similar elastic storage capacity within the fascial tissues of their legs compared with

Chapter 8

their long-legged homo sapiens relatives. This is reflected in the shorter muscle fascicles and thinner tendons in the distal legs of humans compared with non-human primates, making them much better adapted to the elastic storage and release of kinetic energy (Alexander, 1991). The human species' elastic capabilities are not limited to the lower limbs (Figure 8.1). Humans can throw projectiles at much higher speed compared to other primates. Anatomical features that enable elastic energy storage and release in the shoulder contribute to this unique capacity in homo sapiens (Roach *et al.*, 2013).



Figure 8.1

Humans are able to throw projectiles at much higher speed than other primates. However, repeated overuse of this motion can result in injuries in modern athletes. Palaeolithic hunters almost certainly threw far less frequently than modern athletes (Roach *et al.*, 2013).

These discoveries have led to an active revision of long-accepted principles in the field of movement science. According to their previously unquestioned classical model of muscular dynamics, it was assumed that in a muscular joint movement, the skeletal muscle fibers involved shorten. This energy then passes through passive tendons, resulting in the movement (Figure 8.2). This classic form of energy transfer is still true, according to these recent measurements, for slow as well as more rapid movements with a steady limb speed, such as cycling. Here, the muscle fibers actively change their length, while the tendons and aponeuroses scarcely lengthen. The fascial elements mostly fulfil a passive role in this movement orchestration. This is in contrast to oscillatory movement with an elastic spring quality, such as hopping, running or skipping, in which the length of the muscle fibers scarcely changes. While the muscle fibers contract in an almost isometric fashion, they stiffen temporarily without any significant change of their length, the fascial elements function in an elastic way with a movement similar to that of a resonating elastic spring made of stainless steel. It is mainly this lengthening and shortening of the fascial elements that “produces” the actual movement. Think of a catapult: the person putting their muscular energy into the tensioning of the elastic band is like the muscle fibers, which store the kinetic energy in the fascial tissues. Usually the direction of the muscular preparatory work goes into an opposite orientation to the intended explosive power movement. For an archer, this would mean that their muscles pulled backwards on the bow to increase the potential movement energy. When the force is released, it is the collagen (or elastic band) that

Elastic storage and recoil dynamics

propels the object (or arrow) in the intended direction, while the muscular elements step out of the way and take a rest.

Key concept: Elasticity of elastin and collagen fibers

- Early anatomists were of the opinion that tendons did not exhibit elastic properties.
- While today the elasticity of collagenous tissue is beyond any doubt, practitioners and clinicians sometimes mistake elastin fibers to be elastic and collagenous fibers to be non-elastic.
- Elastin and collagen are not more or less elastic; the difference is rather that elastin and collagen

have different quantities of elastic stiffness (quantified by Young's modulus).

- Collagen is about 100 times stiffer than elastin, meaning it can store the same amount of energy while stretching only one hundredth the amount. [Zorn and Hodeck, 2011].

Imagine an archer, who tries to shoot faster by propelling the arrow forward with their muscles. This would be too slow. When properly loaded, collagen can shorten at a much faster speed than any muscle fiber could contract. However, this only works if the loaded collagen tissue has a high elastic storage capacity. Imagine an archer

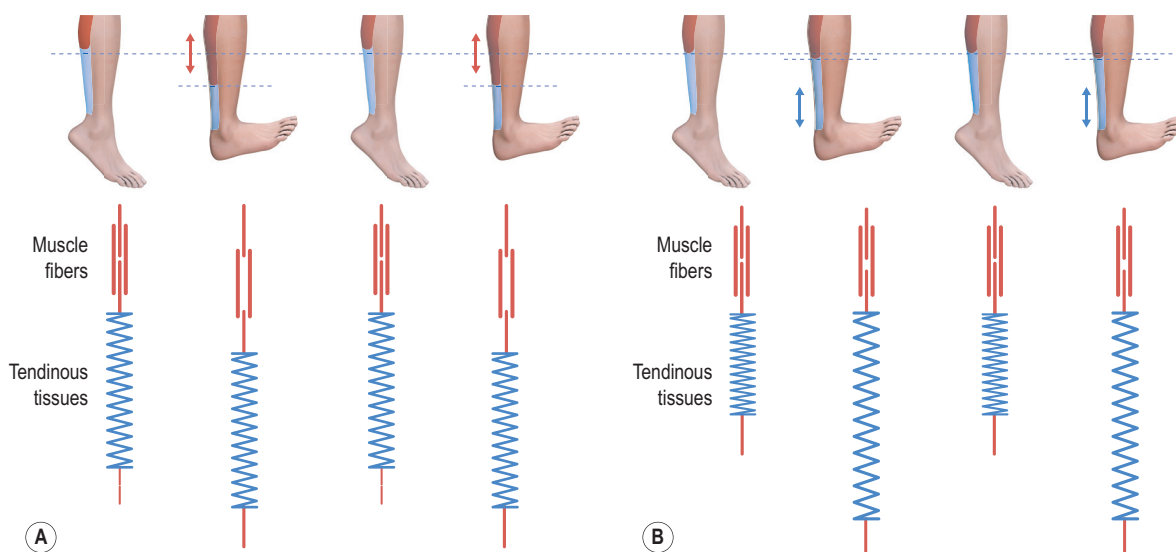


Figure 8.2 Comparison of length changes of muscular and collagenous elements in conventional "muscle training" (A) and in a more fascially oriented movement with elastic recoil properties (B)

The elastic tendinous (or fascial) elements are shown as springs, the muscle fiber as straight lines. Note that during a conventional movement (A), the elastic collagenous elements do not change their length significantly, while the contractile muscle fibers change their length significantly. (B) However, during rhythmically oscillatory movements, like running or hopping, the muscle fibers contract almost isometrically, while the fascial collagenous elements lengthen and shorten like an elastic yo-yo spring.

(Illustration adapted from Kawakami et al. (2002) reproduced with kind permission of www.fascialnet.com).

Chapter 8

trying to shoot with a bow made of concrete and a non-elastic rope.

Key concept: Elastic storage capacity and crimp pattern

- Many tendons can recoil elastically when a stretching force is removed (Benjamin et al., 2008).
- This elastic recoil property seems to be related to the so-called crimp pattern.
- In particular tendons, the fibers are made up of fibrils that have wavy forms called crimps (Shim et al., 2012). Recent animal studies revealed that collagen crimp characteristics are strongly tendon type specific, which indicates that crimp specialization is crucial in the respective mechanical function (Spiesz et al., 2018; Zuskov et al., 2020).
- The stiffness of tendons can vary with age. The higher storage capacity of collagenous tissues in young people is associated with a stronger crimp expression within their collagen fibers. Tailored exercise in older rats has been shown to reintroduce a similar crimp formation as is present in younger ones (Wood et al., 1988).
- Tendon stiffness can also vary with sex: The Achilles tendon in women tends to be less stiff than in men (Kubo et al., 2003).

Resonant frequency: Length and stiffness as crucial factors

In order to understand fascial recoil, try experimenting with a weight that is suspended on an elastic band or stainless steel spring. With eyes open or closed, within a few cycles, most people intuitively figure out the optimal resonant frequency of the system. This is usually achieved when finding a rhythm in which a tiny lift action of the holding finger, starting just a split second

before the turning point, increases the stretch loading of the elastic spring. Interestingly, in a completely elastic system, the ideal resonant frequency depends mostly on two factors: (1) the length of the swinging element and (2) the stiffness of the strained tissues. When trying to dance or bounce on wet sand, most people will find that slower music is more suitable, whereas a more rapid rhythm tends to suit dancing barefoot with the forefeet on a hard dance floor with tightly contracted calf muscles.

Improvement of elastic recoil properties through training

Tissues tend to adapt their architecture to the specific mechanical demands imposed on them. This also holds true for soft connective tissues such as tendons.

While aging alters the mechanical properties, it has been shown that the elastic storage capacity in human tendons can be significantly improved by regularly repeated strong mechanical loading through resistance training (Reeves, 2006).

In addition to resistance training leading to a beneficial adaptation of the tendon, repetitive hopping training has also been shown to improve its mechanical properties (Figure 8.3). A group of physically active elderly men practiced a carefully orchestrated hopping training exercise three times per week for 11 weeks. After a warm-up, the training sets comprised only 15–20 hops with short contact time and as little knee bending as possible. Jumping height and tendon utilization improved (Hoffren-Mikkola et al., 2015).

For the Achilles tendon and aponeuroses, a strain of 4–5% tends to be required for eliciting an adaptational response, while a lower load of 2–3% does not appear to be sufficient

Elastic storage and recoil dynamics



Figure 8.3 Repetitive hopping training can improve mechanical properties and tendon utilization.

A study with elderly active people showed that 15 to 20 hops with short contact time and little knee bending lead to beneficial adaptations.

(Arampatzis et al., 2007). However, the thresholds for adaptational responses appear to be significantly lower for intramuscular fascia than for tendons (Kjaer et al., 2015).

Once the adaptational threshold is reached, it is possible that only 5 or 10 elastic bounces might be required for an adaptational response, while adding many more repetitions tends to have little additional effect (Magnusson et al., 2010).

As described earlier, the rhythm of a movement should also be considered. For rhythmic movements that involve temporarily leaving the ground with both feet, such as running, hopping and skipping, rhythms of between 150 and 170 beats per minute (BPM) usually work best. When trying to find music to support effortless swinging and bouncing movements that do not involve leaving the ground, most people find that a slower rhythm of between 120 and 140 BPM is optimum. But what do you do if the music is too slow or too fast? Besides options such as only choosing every second beat, it may work to change the length of the swinging boy portion and/or to adapt the stiffness of the

loaded fascial tissues by increasing or decreasing the active tonus of some of the attached muscle portions.

Kangaroo hopping versus frog jumping

In elastically oscillating movements, such as hopping in kangaroos, the energy production of the involved muscles is mainly concerned with pacing the best rhythm. This is comparable with the movements of a conductor's baton or the sticks of a slalom skier. It is, therefore, not surprising that the muscles involved when a kangaroo hops generate the same force at all speeds (Kram and Dawson, 1998). Similarly, a study examining the gait of women of the African Luo and Kikuyu tribes carrying loads of up to 20% of their body weight on their heads found that their oxygen consumption was largely independent of the weight, provided they were allowed to walk at a speed comfortable for them. When well-trained British soldiers carrying up to 20% of their body weight on their backpacks were examined, their energy consumption increased in proportion to the

Chapter 8

weight carried. Interestingly, when the African women were asked to walk at an uncomfortably faster or slower pace, they exhibited the same weight-dependent (and probably more muscular-driven) pattern in their energy expenditure as the soldiers (Alexander, 1986; Zorn and Hodeck, 2011).

Compared with these rhythmically oscillating movements, a different orchestration between muscles and elastic tissue properties is at work during single explosive movements. For example, when a frog leaps, sometimes up to 10 m, it also utilizes the elastic recoil properties in its leg fasciae. However, here the contraction speed of the muscle fibers is of prime importance. The orchestration between muscular contractile dynamics and the rebound dynamics of collagenous tissues is similar to casting a long elastic fishing rod. A rapid muscular motion then starts to exert a strong pulling force on the distal limb as well as on related collagenous tissues. While initially some of that force seems “lost” in the elastic compliance of tendons and aponeuroses, these tissues subsequently release the stored muscular energy via a rapidly accelerating motion, the speed of which surpasses the potential maximal muscle contraction velocity by several times. Note that in such “singular” motions, a sense of rhythm or resonant frequencies is not as crucially involved as in oscillatory rhythmic motions. In addition, the magnitude of the initial muscular contraction is of major importance: the more rapidly and strongly the muscles start to contract, the more powerful the resulting elastic recoil action of the fascial tissues will become.

Of course, various complex combinations may exist between these two extremes. As mentioned above, throwing a baseball or a javelin has been shown to exert acceleration speeds that

significantly surpass those of all other primates (Roach *et al.*, 2013). This is partly accomplished by a muscularly driven “stretching backwards” motion (i.e. in the opposite direction to the intended forward throwing motion), in which kinetic energy is stored in various collagenous membranes and tendons. This initial phase simulates some of the hopping kangaroo dynamics, as explored earlier, in which a sense of rhythm and resonant frequencies play important roles. The subsequent phase, consisting of a forward swinging motion, then utilizes similar storage and release dynamics to those found in the powerful frog jump.

Plyometrics: Two different mechanisms

For competitive athletes, plyometric training is considered old hat. The field of plyometrics, also known as jump training, was first introduced to the Western sports scene in the early 1980s. It involves training routines that move the total length of a “muscle-tendon complex” from a state of extension to a shortened state in a rapid or “explosive” way, such as in repetitive jumping. This is also called the “stretch-shortening cycle”. The eccentric contraction of involved muscle fibers during the initial lengthening phase has been shown to play an essential role: the more rapid and powerful these preparatory eccentric contractions are, the more force is subsequently exerted in the final contraction phase. Plyometric training has been shown to increase jumping height, as well as many other athletic performances (Figure 8.4).

The general explanation for the proposed mechanisms is a combination of two factors (Figure 8.5): (1) a change in mechanical properties and (2) an optimization of neural properties (Kubo *et al.*, 2007). While it was unclear to

Elastic storage and recoil dynamics



Figure 8.4

Plyometric training, if done properly, can increase jumping height and induce a gradual remodeling of tendinous tissues towards an increased kinetic storage capacity.

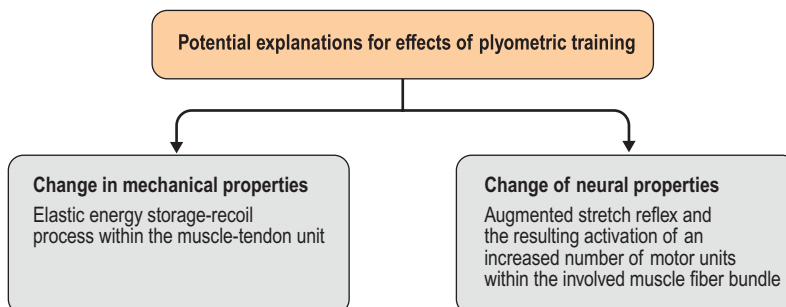


Figure 8.5

Recent investigations revealed that the performance improvement in Plyometric training goes along with a 'lazier' muscle activation in trained jumpers together with an augmented use of passive elastic recoil properties (Fouré, 2011).

what respective degree each of those very different mechanisms are involved in the total outcome, most authors, and athletes, considered the improved neuromuscular orchestration and the assumed augmentation of myogenic contraction power as the dominant player. By contrast, the proposed utilization of passive biomechanical storage and release properties was considered to play a secondary "supporting role" compared with the active muscular contraction component.

Modern ultrasound technology allowed a team of French sports scientists to investigate the relative proportion of those two mechanisms (Fouré

et al., 2011). By measuring length changes in the triceps surae muscles and the Achilles tendon, they observed that plyometric training, consisting of 34 sessions of 1 hour over 14 weeks, resulted in an increased utilization of passive tissue properties and a decreased utilization of active contractile muscular elements. After this systematic training process, the increased acceleration and jumping performance was accompanied by more 'lazy' muscle fibers during the shortening phase but with an increased passive "elastic recoil property" of the spring-loaded elements within the muscle-tendon unit. The authors discussed an increase in cross linkages between collagen

Chapter 8

fibrils as possible explanations for this increase, rather than changes in gross geometrical parameters of the tendons.

If the storage-release mechanism is occurring somewhere within the total muscle-tendon complex, then the question is worth exploring, whether it happens mostly within the collagenous elements (the tendon, epimysium and intramuscular connective tissues), or whether it also occurs within the muscle fibers themselves. Muscle fibers are composed of smaller tubular myofibrils, the basic functional unit of which is called a sarcomere. Several sarcomeres arranged in series constitute a myofibril. The most appealing candidates for elastic energy storage within the sarcomeres are the titin proteins (Linke, 2000). These are the largest known proteins in the human body and they form a more recently discovered third filament system within a sarcomere, apart from the previously known thick (mostly myosin) and thin (mostly actin) filaments. While these unique proteins are capable of impressive elastic recoil properties, their contribution to the stretch-shortening cycle in a rapidly accelerating single movements, such as in jumping or throwing, constituted a matter of dispute. However, a detailed examination of isolated myofibers from frogs revealed that the potential elastic recoil power of these titin proteins tends to be considerably dampened by other intramuscular contraction dynamics. They may, therefore, be prevented from contributing to the elastic recoil properties in a significant manner. The researchers concluded that titin elastic recoil is able to support the active muscular shortening under low load only or during prolonged shortening from greater physiological sarcomere lengths (Minajeva et al., 2002). In other words, with the high loads applied during most plyometric training routines, the intramuscular titin elements seem to play only a minor

role in the passive recoil dynamics of the total muscle-tendon complex. Furthermore, previous studies report that the titin isoforms of human muscles did not change after plyometric training (Pellegrino et al., 2016). These current results suggest that the collagenous (i.e. fascial) tissues are, most likely, providing the main energy storage and release mechanisms that are synonymous with the increased athletic performance achieved using plyometric exercise training.

But what about the potential contribution of the intramuscular titin elements to oscillatory rhythmic recoil movements such as in kangaroo hopping and human running? Here, the length changes of the affected myofascial tissues involve only minor length changes within the muscular sarcomeres, in contrast to the collagenous tendinous elements (Figure 8.2). Since the titin elements are embedded within the sarcomeres, this clearly suggests that most of the elastic recoil dynamics in such movements are achieved within the collagenous elements and not within the muscular titin components. In other words, the training of elastic storage capacities, whether in single motions such as throwing or in oscillatory movements such as jogging, mainly involves an increase of collagenous elastic properties, rather than intramuscular sarcomeric adaptations. This conclusion is in line with a recent examination of the adaptive response of Achilles tendons and patellar tendons in two groups of athletes who use a lot of elastic recoil (ski jumpers and runners) in comparison with other athletes (here, water polo players) and also to sedentary people (Wiesinger et al., 2017). It was found that the energy return in the tendons of the ski jumpers and runners was between 40% and 50% higher in the “bouncy sports athletes” compared with the other groups. The authors therefore concluded that this difference indicates a better

Elastic storage and recoil dynamics

energy conservation in athletes whose tendons mainly work as springs.

Clinical summary

- Fascial tissues are capable of storing and releasing kinetic energy similar to an elastic spring.
- While this catapult mechanism was first examined in a detailed manner in Australian kangaroos, subsequent research revealed that it also plays a major role in human oscillatory movements such as running, hopping or walking.
- Adequate training to improve the elastic capabilities of fascial tissues includes repeated strong mechanical loading through resistance training, repetitive hopping training and rapid accelerating movements.
- The threshold for adaptational responses is significantly lower for intramuscular fascia than for tendons.
- As was examined in the energy efficient gait of some African women, there are significant inter-individual differences as to how much this elastic recoil mechanism is utilized.
- A key factor in the orchestration is adequate timing: the precise matching of the muscular activation with the resonant frequency of the swinging system.

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Chapter 8

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